

## ARTICLE

# A Convergent Multistep Synthesis of Hexaphenylbenzene from Benzaldehyde†

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Hexaphenylbenzene was synthesised from benzaldehyde via a seven-step convergent route, in which tetraphenylcyclopentadienone and diphenylacetylene were prepared by two parallel three-step sequences and coupled in a final Diels–Alder cycloaddition with concomitant loss of CO. The target compound was obtained in 12.98% overall yield, with all intermediates and the final product characterised by melting point, Thin-Layer Chromatography, <sup>1</sup>H NMR, and FTIR spectroscopy.

## Introduction

Hexaphenylbenzene (HPB) is an extraordinary organic molecule with a central benzene ring bearing six peripheral phenyl substituents (fig. 1a), giving it the molecular formula C<sub>42</sub>H<sub>30</sub> and a molecular weight of 534.69 g mol<sup>−1</sup>.<sup>7</sup> Despite its extensive aromatic content, it is not a planar  $\pi$ -system<sup>6</sup>; rather, it is the parent member of the hexaarylbenzene family, a class of sterically crowded aryl frameworks that has attracted continued interest in supramolecular and materials chemistry. Such scaffolds are valuable because their rigid three-dimensional geometry can be tuned extensively through aryl substitution while avoiding the fully planar packing associated with stronger face-to-face  $\pi$ -stacking.<sup>1</sup>

Hexaphenylbenzene itself is a white crystalline solid with a remarkably high melting point of 454–456 °C<sup>7</sup>, a direct consequence of its large molecular structure. The six peripheral phenyl rings create an extensive surface area that maximizes van der Waals dispersion forces between molecules, while the planar aromatic structure enables efficient molecular packing in the solid state.

The chemical interest of hexaphenylbenzene arises from the balance between conjugative stabilization and steric repulsion<sup>6</sup>. A coplanar arrangement of all seven rings would maximize orbital overlap,

but severe ortho–ortho crowding between neighbouring phenyl groups makes that geometry unfavourable.<sup>6</sup> X-ray crystallography showed that the peripheral phenyl rings are rotated by approximately 65° from the plane of the central benzene ring, giving the molecule a six-bladed propeller conformation (fig. 1b).<sup>6</sup>

The first successful synthesis of HPB via the Diels–Alder reaction was performed in the early twentieth century by Dilthey and co-workers in 1933.<sup>8</sup> The formation of HPB from the reaction of tetraphenylcyclopentadienone and diphenylacetylene is a representative example of a Diels–Alder [4+2] cycloaddition reaction followed by thermal cheletropic extrusion of carbon monoxide (Scheme 1).<sup>9</sup> The escape of CO gas also drives the reaction forward according to Le Chatelier's principle, pushing the equilibrium toward more product.

The structure of hexaphenylbenzene was determined by single-crystal X-ray diffraction (XRD) in 1968 by Bart, revealing two polymorphic forms: pyramidal and orthorhombic.<sup>6</sup>

In the pyramidal form (fig. 2b), the six peripheral phenyl rings all tilt to the same side of the central benzene plane, giving the molecule a propeller-like three-dimensional shape that results in an asymmetric, pyramid-like crystal packing.<sup>2</sup> In the orthorhombic form (fig. 2a), the phenyl rings alternate in their tilting — some pointing up and some pointing down — producing a more symmetric, rectangular arrangement in the crystal lattice.<sup>2</sup>

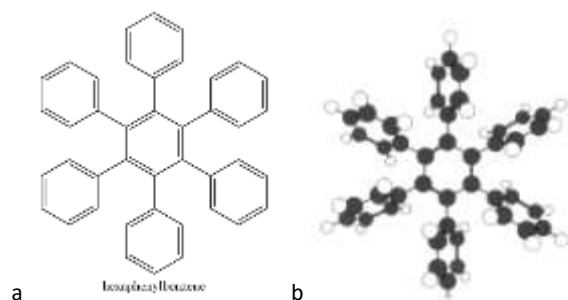
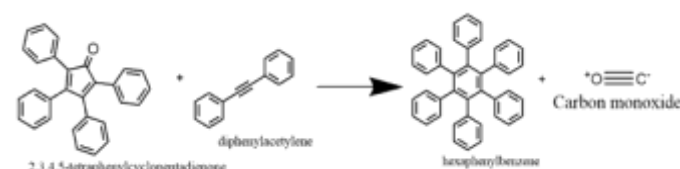


Fig 1 (a) Structural representations of hexaphenylbenzene showing its Skeletal, (b) and six-bladed propeller conformation in which the peripheral phenyl rings are rotated approximately 65° from the central benzene plane.



Scheme 1: Synthesis of hexaphenylbenzene

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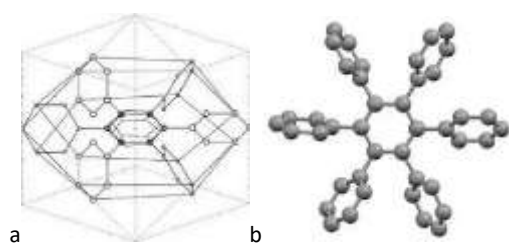


Fig 2: (a) Orthorhombic and (b) pyramidal forms of hexaphenylbenzene

Hexaphenylbenzene has emerged as a highly versatile building block in modern materials chemistry, with several significant applications reported in recent literature. Its enhanced molecular rigidity, arising from the hindered free rotation of each peripheral phenyl ring due to neighbouring electronic repulsion, makes HPB an ideal monomer for the preparation of novel Polymers of Intrinsic Microporosity (PIMs) via condensation polymerization (fig. 3).<sup>1,10</sup> The resulting HPB-based polymeric membranes have demonstrated remarkable efficiency and selectivity in gas separation applications, with the additional advantage that the HPB core can be decorated with different substituents to fine-tune selectivity for various gas pairs.<sup>1</sup> Of particular relevance to current global challenges, HPB-based PIMs with varying carbon-to-nitrogen (C/N) ratios have been designed to enhance CO<sub>2</sub> selectivity toward N<sub>2</sub>, positioning them as promising candidates for post combustion carbon capture from industrial flue gas.<sup>1</sup> Furthermore, the outstanding thermal stability inherent to HPB-PIMs has drawn attention to their potential as flame-retardant materials, where research has shown that systematically lowering the C/N ratio within the polymer framework correlates directly with improved flame-retardant performance.<sup>1</sup>

HPB serves as a key building block for the fabrication of atomically precise covalent nanostructures for applications in molecular electronics. A brominated HPB derivative, Br<sub>2</sub>-HPB, is used as a precursor for thermally induced on-surface polymerisation on an Au(111) gold surface.<sup>4</sup> When Br<sub>2</sub>-HPB is heated on the gold surface, it undergoes debromination and polymerisation, forming one-dimensional HPB polymer chains.<sup>4</sup> Br<sub>2</sub>-HPB molecules can organise into self-assembled layers on gold surfaces, making them highly promising for the bottom-up fabrication of precisely designed nanoscale materials and devices.<sup>4</sup>

HPB derivatives have further demonstrated significant utility in the construction of metallacages (fig. 4)—three-dimensional hollow cage-like structures formed through the multicomponent self-assembly of metal ions and organic ligands—via a process driven by metal coordination.<sup>3</sup> The resulting HPB-based metallacages exhibit deep blue emission in both solution and solid state.<sup>3</sup> These light-emitting metallacages hold considerable promise across a range of applications including optoelectronic devices, chemical and biological sensors, and imaging agents.<sup>3</sup> The HPB metallacages can act as energy donors in light-harvesting systems, where they transfer energy to an acceptor molecule such as a naphthalimide derivative.<sup>3</sup>

HPB serves as the most straightforward and efficient precursor for synthesizing hexa-peri-hexabenzocoronene (HBC) — a prominent PAH known as "super benzene" — through an intramolecular dehydrocyclization process.<sup>5,12</sup> HBC is described as a fragment of graphene, so using HPB to make HBC connects HPB to the bottom-up synthesis of graphene/nanographene structures, opening pathways toward the fabrication of advanced nanoscale materials and molecular electronic devices (fig. 5).<sup>5</sup>

HPB is commonly used as a sterically hindered fluorophoric core in organic luminogens.<sup>5</sup> Its rotor-like molecular geometry favours inefficiently packed crystal structures,<sup>13</sup> which restricts the motion of the phenyl substituents and gives rise to aggregation-induced emission (AIE).<sup>5,13</sup> Due to its inability to form close  $\pi\cdots\pi$  stacks, HPB-based luminogens emit short-wavelength fluorescence in the violet

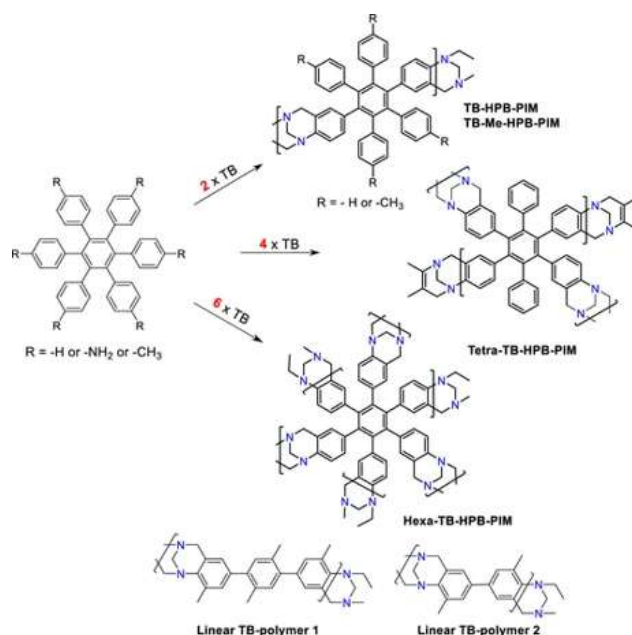


Fig 3: Synthesis scheme illustrating the preparation of a series of HPB-based PIMs with varying degrees of functionalization, yielding networked polymers of increasing complexity alongside two linear reference polymers.

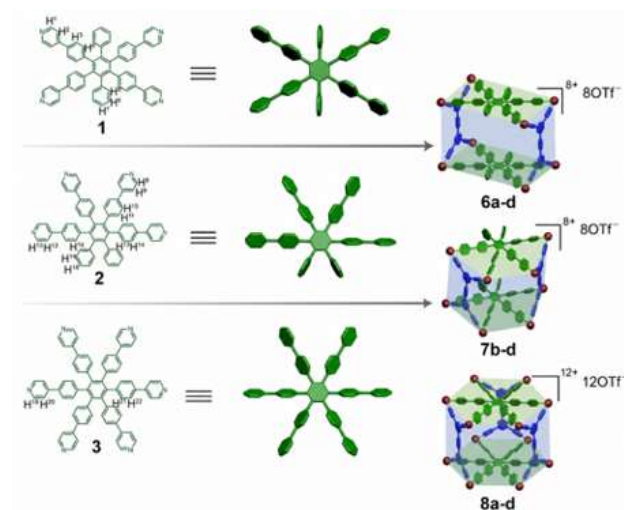


Fig 4: Self-assembly of three HPB derivatives into their corresponding three-dimensional metallacages (6a-d, 7b-d, and 8a-d).

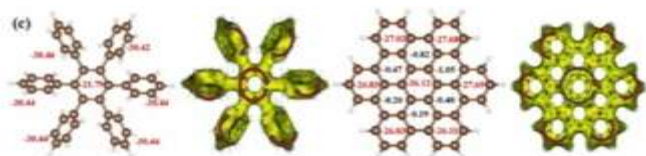


Fig 5: Molecular structures and aromaticity maps of HPB (left) and HBC (right), showing how HPB transforms into the fully aromatic HBC upon dehydrocyclization.

or blue region, making them relevant to optoelectronic devices, Viscosity probes, Bioimaging, Cancer therapeutics, Batteries.<sup>5</sup>

In this study, the Diels-Alder reaction is employed as the synthetic route to hexaphenylbenzene, with the aim of investigating its use in forming six-membered aromatic rings via decarbonylation and aromatization of a bicyclic intermediate. Following a convergent synthetic sequence, each product is characterised using melting point determination, IR, and <sup>1</sup>H NMR spectroscopy, collectively providing a comprehensive examination of the properties of hexaphenylbenzene as the final synthetic target.

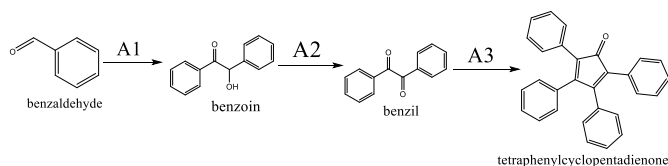
## Results and Discussion

The convergent synthesis of hexaphenylbenzene from benzaldehyde was accomplished over seven sequential reactions, comprising two parallel synthetic pathways (Series A and Series B) (Scheme 2 and 3) that converged in a final Diels-Alder cycloaddition (Scheme 4).

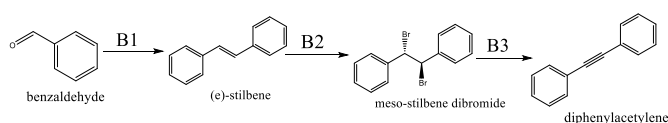
### Reaction A1:

Treatment of benzaldehyde with catalytic sodium cyanide in refluxing 95% ethanol afforded benzoin as an off-white crystalline solid in 47.91% yield following recrystallisation (Scheme 5). The moderate yield is consistent with the reversible nature of the benzoin condensation (Scheme 6) and is typical for this transformation at the microscale, where the equilibrium nature of each mechanistic step limits the conversion attainable in a single reaction.

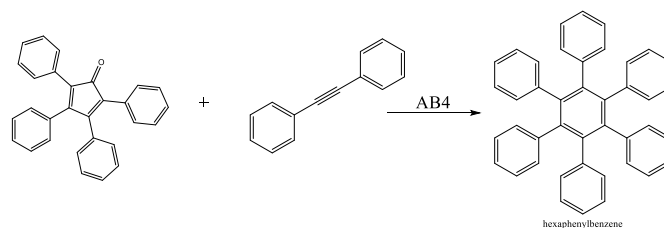
The observed melting point of 196.2 °C was substantially higher than the reported literature value of 137 °C.<sup>11</sup> As impurities almost invariably depress and broaden a melting point rather than elevate it, this discrepancy is most likely attributable to an excessive ramp rate during the determination, in which the apparatus temperature overshoot the true melting point before the transition was registered.



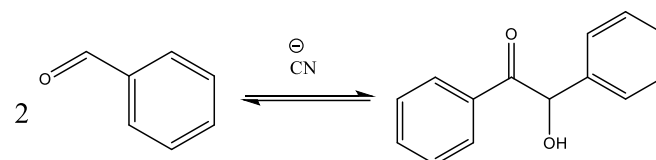
Scheme 2: The "A" series of reactions



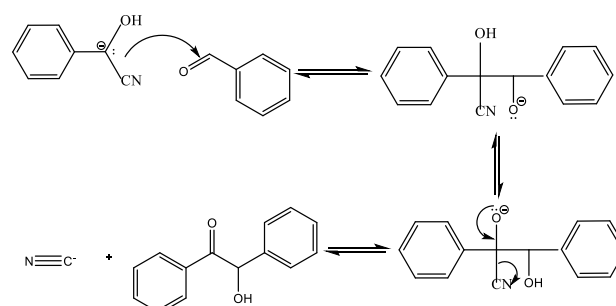
Scheme 3: The "B" series of reactions



Scheme 4: Synthesis of hexaphenylbenzene



Scheme 5: Reaction A1



Scheme 6: Benzoin condensation mechanism

Re-determination at a reduced ramp rate of 1–2 °C min<sup>-1</sup> in the vicinity of the expected melting point would be expected to yield a value in closer agreement with the literature.

The FTIR spectrum (see ESI<sup>†</sup>) of the product provided strong evidence for the successful formation of benzoin. A broad absorption centred at 3376 cm<sup>-1</sup> was assigned to the O–H stretching vibration of the newly introduced secondary alcohol, a functional group absent in the benzaldehyde starting material and diagnostic of the condensation product. A strong band at 1677 cm<sup>-1</sup> was assigned to the carbonyl stretch of the aryl ketone; the depression of this frequency relative to a typical unconjugated ketone (~1715 cm<sup>-1</sup>) reflects conjugation with the adjacent phenyl ring. The absence of the diagnostic aldehyde C–H stretches (typically 2720–2820 cm<sup>-1</sup>) confirmed that the benzaldehyde starting material had been consumed.

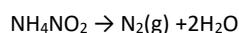
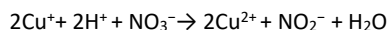
The <sup>1</sup>H NMR spectrum (see ESI<sup>†</sup>) recorded in CDCl<sub>3</sub> was likewise fully consistent with the proposed structure. The aromatic region (δ 7.32–8.05 ppm) integrated for approximately ten protons, corresponding to the two phenyl rings. A diagnostic singlet at δ 5.97 ppm, integrating for one proton, was assigned to the benzylic methine proton; this resonance appears in this characteristic region only when a CH is flanked simultaneously by a hydroxyl group and a carbonyl, and thus serves as definitive evidence that dimerization has occurred. A broad signal at δ 4.14 ppm integrating for one proton was attributed to the exchangeable hydroxyl proton, the breadth of which is characteristic of an O–H undergoing slow exchange on the NMR timescale.

Thin-layer chromatography (silica gel, dichloromethane mobile phase) was used to assess the purity of the recrystallised product and the composition of the mother liquor (filtrate). With a solvent front

of 4.0 cm, the recrystallised benzoin appeared as a single spot at  $R_f$  0.63, while the filtrate displayed a single spot at  $R_f$  0.50. The appearance of only one spot in each lane indicated the absence of significant impurities such as unreacted benzaldehyde, which would be expected to migrate at a higher  $R_f$  than benzoin. The lower  $R_f$  of the filtrate spot suggests it contained a slightly more polar species than benzoin itself, possibly a trace oxidation product formed during workup, which was effectively excluded from the crystalline product during recrystallisation. This observation is consistent with the purifying role of the crystallisation step.

#### Reaction A2:

Catalytic oxidation of benzoin using copper(II) acetate in the presence of ammonium nitrate in glacial acetic acid afforded benzil (Scheme 7) as bright yellow crystals in 70.92% yield following column chromatography. The improved yield relative to the benzoin condensation reflects the irreversible nature of this transformation, in which the  $\alpha$ -hydroxyketone is cleanly oxidised to the  $\alpha$ -diketone with concurrent evolution of nitrogen gas.

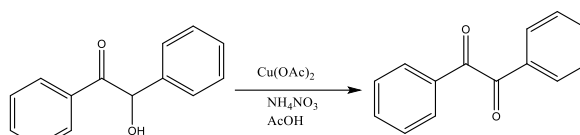


The observed melting point of 91.3 °C was in close agreement with the literature value of 94–95 °C<sup>14</sup>, with the modest 3 °C depression most likely attributable to either a trace residual impurity or a slightly elevated ramp rate. The narrow deviation, in contrast to the substantial discrepancy observed for benzoin (A1), supports the assignment of the product as benzil and indicates that the column chromatography effectively purified the material.

The FTIR spectrum (see ESI<sup>†</sup>) provided particularly compelling evidence for the success of the oxidation. The broad O–H stretching absorption at 3376 cm<sup>-1</sup> that was prominent in the benzoin precursor was completely absent in the product spectrum, confirming the removal of the hydroxyl functionality during oxidation. The carbonyl stretching frequency was observed at 1657 cm<sup>-1</sup>, representing a further depression from the 1677 cm<sup>-1</sup> value recorded for benzoin. This downward shift is consistent with the additional conjugation present in the  $\alpha$ -diketone system of benzil, where the two carbonyl groups are directly bonded to one another, allowing electronic delocalisation that weakens each individual C=O bond.

The <sup>1</sup>H NMR spectrum (see ESI<sup>†</sup>) recorded in CDCl<sub>3</sub> provided definitive corroboration of the structural change. The aromatic region ( $\delta$  7.27–8.21 ppm) integrated for ten protons in total, consistent with the two phenyl rings of benzil. Most diagnostically, the singlet at  $\delta$  5.97 ppm corresponding to the benzylic methine of benzoin, and the broad O–H signal at  $\delta$  4.14 ppm, were both entirely absent from the spectrum. The loss of these two resonances provides unequivocal evidence that the central CH(OH) unit of benzoin has been oxidised to the C=O of benzil, completing the conversion of the  $\alpha$ -hydroxyketone to the  $\alpha$ -diketone.

Prior to column chromatography, thin-layer chromatography (silica gel, dichloromethane mobile phase) was performed on the crude reaction mixture to assess whether purification was required. With a solvent front of 4.0 cm, the crude product lane displayed two spots: a major spot at  $R_f$  0.75 corresponding to benzil, and a less intense lower spot at  $R_f$  0.70 attributable to unreacted benzoin starting material. The mother liquor lane likewise contained both compounds, displaying an elongated streak in the region of  $R_f \approx 0.50$ . While the absolute  $R_f$  values exceeded the literature values of 0.62



Scheme 7: Reaction A2

and 0.33 reported for benzil and benzoin respectively in dichloromethane, the relative ordering of the two spots — with benzil migrating further than benzoin owing to the absence of the polar hydroxyl group — was consistent with the expected chromatographic behaviour. The presence of two spots in the crude material confirmed that the Cu(II)-mediated oxidation had not proceeded to completion, justifying the use of column chromatography to separate the desired  $\alpha$ -diketone product from the unreacted  $\alpha$ -hydroxyketone starting material. The subsequent column purification yielded benzil sufficiently pure for spectroscopic characterisation and use in the following step.

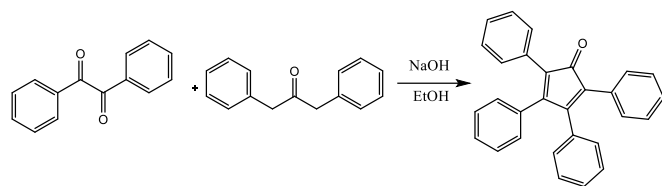
#### Reaction A3:

Base-mediated aldol condensation of benzil with 1,3-diphenylacetone in ethanolic potassium hydroxide afforded tetraphenylcyclopentadienone (TPCPD) (Scheme 8) as a deep purple-black crystalline solid in 69.90% yield. The dramatic colour change from the bright yellow benzil starting material to the intensely coloured product provided immediate visual evidence of the formation of an extensively conjugated  $\pi$ -system. This colour arises from the cross-conjugated cyclopentadienone framework, in which the carbonyl group is in direct conjugation with two carbon–carbon double bonds and four pendant phenyl substituents; the highly delocalised electronic structure permits absorption of light in the visible region.

The observed melting point of 219 °C was in excellent agreement with the literature value of 219–220 °C<sup>15</sup>, with no discernible deviation. This close correspondence provides strong evidence of high product purity and confirms that no further purification of the crystalline material was required prior to spectroscopic characterisation or its use in the final Diels–Alder step.

The FTIR spectrum (see ESI<sup>†</sup>) strongly supported the assigned structure. The carbonyl stretching absorption was observed at 1646 cm<sup>-1</sup>, representing a further downward shift relative to benzil (1657 cm<sup>-1</sup>) and benzoin (1677 cm<sup>-1</sup>). This progressive lowering of the C=O frequency through the synthetic series directly reflects the increasing extent of conjugation: while benzoin's carbonyl is conjugated only with a single phenyl ring, benzil's  $\alpha$ -diketone introduces additional conjugation, and tetraphenylcyclopentadienone's cross-conjugated dienone places the C=O in conjugation with two endocyclic C=C bonds as well as the four phenyl substituents. The resulting  $\pi$ -electron delocalisation weakens the C=O bond and depresses its stretching frequency, an observation that is diagnostic of the cyclopentadienone framework.

The <sup>1</sup>H NMR spectrum (see ESI<sup>†</sup>) recorded in CDCl<sub>3</sub> was striking in its simplicity, displaying signals exclusively in the aromatic region ( $\delta$  6.82–7.75 ppm) with no resonances elsewhere in the spectrum. The integration over this entire range corresponded to twenty protons, in exact agreement with the expected count for the four phenyl substituents of TCPD (4  $\times$  5H = 20H). The complete absence



Scheme 8: Reaction A3

of any aliphatic signals — particularly the lack of  $\alpha$ -methylene resonances near  $\delta$  3.5 ppm that would arise from unreacted 1,3-diphenylacetone, and the lack of any benzylic methine signal near  $\delta$  5.97 ppm characteristic of residual benzoin — provided unambiguous evidence that the double aldol condensation had proceeded to completion, with both  $\alpha$ -hydrogens of the diphenylacetone removed via condensation and dehydration to install the two C=C bonds of the cyclopentadienone ring.

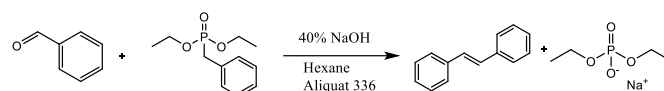
Taken together, the colour, melting point, IR and NMR data unambiguously confirm the successful synthesis of tetraphenylcyclopentadienone. With completion of this step, the A series of the synthesis was concluded; the overall yield of tetraphenylcyclopentadienone from benzaldehyde, calculated over the three sequential steps (A1: 47.91%, A2: 70.92%, A3: 69.90%), was 23.75% (see ESI<sup>†</sup>). The product was reserved for the final Diels–Alder cycloaddition with diphenylacetylene (Experiment AB4).

#### Reaction B1:

The Horner–Wadsworth–Emmons modification of the Wittig reaction was employed to couple benzaldehyde with diethyl benzylphosphonate, using 40% aqueous sodium hydroxide and Aliquat 336 (tricaprylmethylammonium chloride) as a phase-transfer catalyst in a biphasic petroleum spirits/aqueous reaction medium (Scheme 9).

The reaction proceeded with consumption of diethyl benzylphosphonate as the limiting reagent (5.57 mmol) and afforded (E)-stilbene as colourless crystals in 12.05% yield following extractive workup and recrystallisation from 95% ethanol. While this yield is modest, the observed melting point of 121.8 °C was in excellent agreement with the reported literature value of 122–124 °C<sup>16</sup>, indicating that despite the small quantity of material isolated, the recrystallised product was of high purity.

The disparity between the low yield and the high spectroscopic purity is most plausibly attributed to mechanical losses during the multi-step workup (which involved phase separation, multiple aqueous washes, drying over anhydrous sodium sulfate, and rotary evaporation) compounded by an aggressive recrystallisation in which a substantial portion of the product remained dissolved in the mother liquor. The viscous nature of the Aliquat 336 phase-transfer catalyst may also have hindered efficient mixing of the biphasic reaction medium, and incomplete conversion in the alkene-forming step cannot be excluded. Future efforts to improve the yield could focus on more vigorous stirring during the reaction, extended reaction times to ensure complete conversion of the phosphonate, and a less aggressive recrystallisation protocol involving collection of multiple smaller crops.



Scheme 9: Reaction B1

The FTIR spectrum (see ESI<sup>†</sup>) provided strong evidence for the successful formation of (E)-stilbene. The most diagnostically important feature was the strong absorption at 961 cm<sup>-1</sup>, characteristic of the out-of-plane C–H bending vibration of a trans-disubstituted alkene. This band falls within the well-established 960–980 cm<sup>-1</sup> region associated with E-configured C=C double bonds, and its presence, together with the absence of any corresponding cis-alkene absorption near 690–710 cm<sup>-1</sup> that could be unambiguously attributed to a Z-isomer, confirmed the E-stereochemistry of the olefination product.

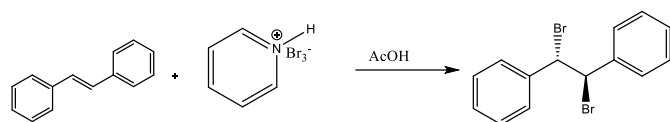
The <sup>1</sup>H NMR spectrum (see ESI<sup>†</sup>) recorded in CDCl<sub>3</sub> corroborated the structural assignment. A multiplet in the aromatic region ( $\delta$  7.40–7.72 ppm) integrated for ten protons, corresponding to the two equivalent phenyl rings of (E)-stilbene, while a signal at  $\delta$  7.15 ppm integrated for the remaining two protons of the molecule, assigned to the two equivalent olefinic protons of the trans-configured C=C bond.

The molecular symmetry of (E)-stilbene renders the two vinylic protons chemically equivalent, accounting for their appearance as a single resonance rather than a coupled pair. The total proton count of twelve corresponded exactly to the molecular formula of (E)-stilbene (C<sub>14</sub>H<sub>12</sub>).

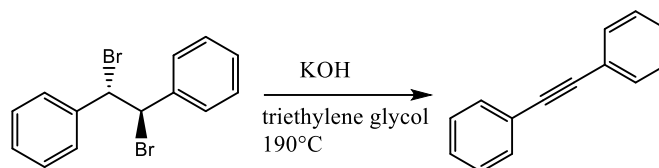
Thin-layer chromatography (silica gel, dichloromethane mobile phase) was performed to assess the composition of the recrystallised product and the mother liquor (filtrate). The product lane displayed a single spot near the solvent front ( $R_f$ =1.0) consistent with the highly non-polar nature of (E)-stilbene, which lacks any polar functional groups and therefore interacts only weakly with the silica stationary phase, traversing the plate almost in step with the dichloromethane eluent. The appearance of only one spot in the product lane confirmed the absence of detectable impurities in the recrystallised material. In contrast, the filtrate lane displayed multiple spots distributed across the length of the plate, indicating that the mother liquor retained both dissolved (E)-stilbene that did not crystallise out and additional more polar species, most likely residual diethyl benzylphosphonate, partially hydrolysed phosphonate byproducts, and trace unreacted benzaldehyde. The successful separation of these impurities into the filtrate during recrystallisation accounts for both the low isolated yield and the high purity of the crystalline product evidenced by the melting point and spectroscopic data.

#### Reaction B2:

Electrophilic bromination of (E)-stilbene (Scheme 10) was performed using pyridinium bromide perbromide as a solid, easily handled source of bromine, in refluxing glacial acetic acid at 130–140 °C. Pyridinium bromide perbromide served as the limiting reagent (32 mmol against 90 mmol of (E)-stilbene), and the reaction afforded meso-stilbene dibromide as white crystals in 8.98% yield following filtration and washing with cold water and cold acetone. Despite the low isolated yield, the observed melting point of 238 °C was in close agreement with the literature value of 241 °C<sup>17</sup>, supporting the high



Scheme 10: Reaction B2



Scheme 11: Reaction B3

purity of the recrystallised product. The narrow  $\sim 3$  °C depression relative to the literature value is consistent with the inherent decomposition behaviour of this compound at its melting point and does not necessarily indicate the presence of significant impurities.

The low yield warrants critical discussion. Several contributing factors are likely: the limited amount of (E)-stilbene available from the previous step (B1 gave only 12.05% yield), mechanical losses at the microscale during the multi-stage filtration and washing protocol, and the aggressive nature of the cold acetone washes specified in the procedure (which were performed "until the product was no longer brownish in colour"). These washes, while necessary to remove the orange pyridinium byproducts, also dissolve appreciable quantities of the crystalline product. Additionally, partial decomposition or loss of HBr cannot be excluded under the elevated reaction temperatures employed. Future work could improve the isolated yield by employing a larger-scale reaction to dilute mechanical losses, by using a finer-pored or sintered-glass funnel to minimise loss of small crystallites during filtration, and by reducing the aggressiveness of the washing protocol — accepting a slightly off-white product would likely substantially improve material recovery.

The FTIR spectrum (see ESI<sup>†</sup>) provided strong evidence for the successful bromination. The most diagnostic feature was the complete disappearance of the strong absorption at  $961\text{ cm}^{-1}$  that had been prominent in the (E)-stilbene starting material; this band corresponds to the out-of-plane C–H bending vibration of the trans-disubstituted alkene, and its loss confirms that the C=C double bond has been consumed by the bromine addition. The presence of several absorptions in the  $500\text{--}700\text{ cm}^{-1}$  region (notably at  $568$ ,  $542$  and  $523\text{ cm}^{-1}$ ) is consistent with newly introduced C–Br stretching modes, providing direct spectroscopic evidence for the incorporation of bromine atoms into the molecular framework.

The  $^1\text{H}$  NMR spectrum (see ESI<sup>†</sup>) recorded in  $\text{CDCl}_3$  provided definitive corroboration of the structural and stereochemical outcome. A multiplet in the aromatic region ( $\delta$  7.16–7.83 ppm) integrated for ten protons, corresponding to the two phenyl rings of the dibromide, while a singlet at  $\delta$  5.46–5.63 ppm integrated for the two CHBr methine protons. The singlet multiplicity of this latter resonance is mechanistically significant: it arises from the magnetic equivalence of the two methine protons, which is a consequence of the internal mirror symmetry of the meso diastereomer. Crucially, the diagnostic vinylic singlet of (E)-stilbene near  $\delta$  7.15 ppm was absent, providing further confirmation that the alkene starting material had been fully consumed.

### Reaction B3:

The synthesis of diphenylacetylene (Scheme 11) was accomplished by a double dehydrohalogenation of meso-stilbene dibromide using potassium hydroxide in triethylene glycol at  $190\text{--}195$  °C. The reaction proceeded with meso-stilbene dibromide as the limiting reagent (1.2 mmol) and a large molar excess of KOH to ensure complete

elimination at both carbon centres. Following aqueous quench, vacuum filtration, and washing with cold 70% ethanol, diphenylacetylene was obtained as a tan crystalline solid in 90.52% yield without the need for recrystallisation or chromatographic purification. The observed melting point of  $60.4$  °C was in excellent agreement with the literature value of  $60\text{--}62$  °C<sup>18</sup>, confirming the high purity of the isolated product.

The high yield obtained at this step contrasts sharply with the modest yields recorded for the preceding B-series transformations (12.05% for B1 and 8.98% for B2) and is attributable to several converging factors. The double elimination is thermodynamically favourable owing to the formation of the strong  $\text{C}\equiv\text{C}$  bond and the loss of HBr as a stable inorganic salt. The use of a large excess of potassium hydroxide ensured rapid deprotonation at both stages of the reaction, and the high boiling point of triethylene glycol permitted operation at temperatures well above  $180$  °C — a requirement for the second elimination, which proceeds by a higher-energy syn pathway and is therefore the rate-limiting step in the overall sequence. Equally significant is the simplicity of the workup: the product was isolated directly by precipitation from the aqueous-diluted reaction mixture, eliminating the multiple solvent transfers, drying steps, and recrystallisation losses that compromised the yields of B1 and B2.

The FTIR spectrum (see ESI<sup>†</sup>) of the product provided strong, if initially counter-intuitive, evidence for the formation of diphenylacetylene. The most diagnostically significant feature of the spectrum is in fact the absence of any absorption in the  $2100\text{--}2260\text{ cm}^{-1}$  region where the  $\text{C}\equiv\text{C}$  stretch of an internal alkyne would conventionally be expected. This apparent omission is fully consistent with the structure of diphenylacetylene: the molecule possesses an inversion centre (see ESI<sup>†</sup>) along the  $\text{C}\equiv\text{C}$  axis, and the symmetric  $\text{C}\equiv\text{C}$  stretch is therefore IR-inactive by the mutual exclusion principle (see ESI<sup>†</sup>), although it remains strongly Raman-active. The absence of a sharp band at approximately  $3300\text{ cm}^{-1}$  further confirms the disubstituted nature of the alkyne, ruling out any terminal  $\equiv\text{C}\text{--H}$  functionality. Aromatic C–H stretching was observed at  $3062\text{ cm}^{-1}$ , with aromatic ring C=C stretches at  $1598$ ,  $1491$ , and  $1441\text{ cm}^{-1}$ . The pair of strong out-of-plane bending modes at  $753$  and  $687\text{ cm}^{-1}$  — the classic fingerprint of a monosubstituted phenyl ring — confirmed that each aromatic ring in the molecule bears only a single substituent. Critically, the spectrum displayed no alkene C=C stretch near  $1640\text{ cm}^{-1}$ , ruling out residual (E)-stilbene starting material or any retained bromoalkene intermediate, and no C–Br absorption in the  $500\text{--}650\text{ cm}^{-1}$  region, confirming that both carbon–bromine bonds of the meso-stilbene dibromide had been cleaved.

The  $^1\text{H}$  NMR spectrum (see ESI<sup>†</sup>) recorded in  $\text{CDCl}_3$  provided unambiguous confirmation of the molecular structure. The entire



spectrum consisted of a single aromatic multiplet between  $\delta$  6.95 and 7.92 ppm, integrating for ten protons and accounting for all of the hydrogen atoms in the molecule. The two phenyl rings are rendered chemically equivalent by the  $C_2$  symmetry of the molecule about the  $C\equiv C$  axis, and no other resonances were observed at any chemical shift. Of particular diagnostic value was the absence of any signal in the regions that would have indicated unreacted starting material or partial-conversion product: no benzylic CHBr signal was observed near  $\delta$  5.4 ppm, confirming that the meso-stilbene dibromide had been fully consumed, and no vinyl signal was observed in the  $\delta$  6.0–7.0 ppm region, confirming that the intermediate (E)- $\alpha$ -bromostilbene formed after the first elimination had itself been carried forward to the alkyne by the second elimination rather than persisting as a contaminant. The integration of exactly ten aromatic protons with no other signals matched the molecular formula  $C_{14}H_{10}$  of diphenylacetylene precisely.

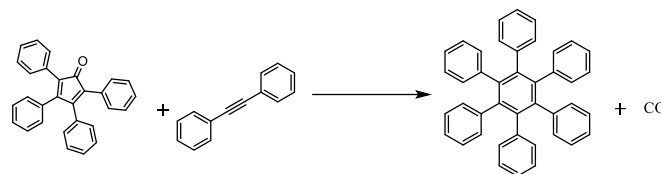
Tracing the three steps of the B-series branch gives an overall yield of 0.98% ( $0.1205 \times 0.08983 \times 0.9052$ ) from benzaldehyde. Although modest in absolute terms, this figure reflects the cumulative impact of the lower yields obtained at B1 and B2, both of which were dominated by material losses during the multi-step extractive workups and aggressive recrystallisations performed in pursuit of high product purity. The strong B3 yield of 90.52% was, by contrast, the highest single-step yield in the entire B branch, and the diphenylacetylene obtained at this stage was of sufficient quality to be used directly in the final convergent Diels–Alder coupling without further purification.

#### Reaction AB4:

The convergent synthesis was completed by combining tetraphenylcyclopentadienone (prepared in A3) and diphenylacetylene (prepared in B3) in a Diels–Alder cycloaddition conducted in high-boiling silicone oil at temperatures above 250 °C. (Scheme 12)

Tetraphenylcyclopentadienone was the limiting reagent (0.26 mmol) and diphenylacetylene was used excess to drive the cycloaddition to completion and to ensure that any residual unreacted alkyne could be removed in the workup, where the marked solubility difference between diphenylacetylene and hexaphenylbenzene permits selective extraction with petroleum spirits (see ESI<sup>†</sup>). The reaction proceeds in two stages: First, diphenylacetylene adds across the diene system of tetraphenylcyclopentadienone in a Diels–Alder cycloaddition to form a bicyclic intermediate that still contains the original C=O group as a one-carbon bridge. At the high reaction temperature, this intermediate then loses carbon monoxide gas, which allows the six-membered ring left behind to become aromatic, giving hexaphenylbenzene. Following dilution with petroleum spirits, vacuum filtration, three petroleum-spirit washes, and two cold toluene washes, hexaphenylbenzene was obtained as an off-white crystalline solid in 54.67% yield.

A melting point was not determined for this product. Hexaphenylbenzene is known to melt without decomposition at  $\sim 465$  °C<sup>7</sup> — a temperature well beyond the safe operating range of the standard melting-point apparatus and thermometer available in the teaching laboratory.



Scheme 12: Reaction AB4

The FTIR spectrum (see ESI<sup>†</sup>) of the product provided the single most diagnostically valuable piece of evidence for the successful completion of the cascade. The carbonyl stretching band of the tetraphenylcyclopentadienone starting material, which appears as a strong absorption at  $\sim 1646$   $cm^{-1}$  for the cyclopentadienone C=O group, was entirely absent from the product spectrum. This absence is not merely consistent with the disappearance of starting material; it positively demonstrates that the initial Diels–Alder cycloaddition was followed by the reverse-cycloaddition in which the bridging carbonyl was expelled as CO. Had the reaction stalled at the bicyclic adduct stage, the C=O group of that intermediate would still have been present, and a strong carbonyl absorption would still have been observed. The complete loss of any band in the 1700–1720  $cm^{-1}$  region therefore confirms full aromatisation. The aromatic C–H stretching modes observed at 3054 and 3023  $cm^{-1}$ , the moderate aromatic C=C stretches at 1598, 1576, 1494, and 1440  $cm^{-1}$ , and most particularly the diagnostic pair of out-of-plane C–H bending vibrations at 729 and 694  $cm^{-1}$  — the classic two-band signature of a monosubstituted benzene ring — together confirm the presence of monosubstituted phenyl rings in the molecule.

The <sup>1</sup>H NMR spectrum (see ESI<sup>†</sup>) recorded in CDCl<sub>3</sub> provided unambiguous structural confirmation. The entire spectrum consisted of a single broad aromatic multiplet between  $\delta$  6.20 and 7.62 ppm, integrating for exactly thirty protons and accounting for the entire proton content of the molecule. The integration matches the molecular formula  $C_{42}H_{30}$  precisely: six equivalent monosubstituted outer phenyl rings each contributing five protons, with the hexasubstituted central ring contributing no protons of its own and thus being invisible to the technique. No signals were observed in the aliphatic region, ruling out residual 1,3-diphenylacetone, partial-conversion saturated by-products, or solvent contamination; and no signals corresponding to the intermediate bicyclic Diels–Alder adduct (which would retain non-aromatic  $sp^3$  centres) were observed.

The yield of 54.67% is moderate by comparison with the standout B3 result but lies within the literature range for this transformation conducted at microscale. The principal factors contributing to material loss are the reversibility of the initial Diels–Alder step (partly offset by the excess alkyne used and by the escape of CO gas, which prevents the reaction reversing), some risk of localised thermal decomposition at the temperatures required, and accumulated transfer and solubility losses during the five sequential wash steps applied during the workup.

This step completes the seven-step convergent synthesis from benzaldehyde to hexaphenylbenzene. The overall yield to the

tetraphenylcyclopentadienone through the A-series pathway was 23.75% ( $0.4791 \times 0.7092 \times 0.6990$ ), and the overall yield to the diphenylacetylene through the B-series pathway was 0.98% ( $0.1205 \times 0.08983 \times 0.9052$ ). Because tetraphenylcyclopentadienone was the limiting reagent in the final coupling, the overall yield from benzaldehyde to hexaphenylbenzene is determined by the A-series pathway combined with the AB4 step: 12.98% ( $0.2375 \times 0.5467$ ). An overall yield of approximately 13% through seven sequential transformations is a respectable outcome for a multistep microscale synthesis involving two parallel three-step branches and a final convergent coupling.

## Conclusion

The convergent seven-step synthesis of hexaphenylbenzene from benzaldehyde was completed successfully. Series A of reactions, proceeded with moderate-to-good yields, delivering tetraphenylcyclopentadienone in 23.75% overall yield from benzaldehyde. Series B of reactions was less efficient overall (0.98% to diphenylacetylene), with the strong B3 yield (90.52%) unable to recover the substantial mechanical losses incurred in the earlier extractive workups and aggressive recrystallisations. The two branches converged in a final Diels–Alder cycloaddition with loss of CO to give hexaphenylbenzene in 54.67% yield at AB4, corresponding to an overall yield of 12.98% from benzaldehyde via the limiting A-series pathway. Across all seven transformations, diagnostic spectroscopic features confirmed both the formation of the desired product and the consumption of relevant starting materials, most notably the complete absence of the cyclopentadienone C=O stretch in the AB4 product spectrum, which confirmed that the Diels–Alder cycloaddition had been followed by full decarbonylation and aromatisation to give the target compound. The stated aim — to construct hexaphenylbenzene by a multistep convergent route and to characterise each intermediate at every stage — was therefore fully met.

Looking beyond established uses, the locked propeller shape of hexaphenylbenzene — in which rotation of any one outer phenyl ring is mechanically linked to its two neighbours through their *ortho* hydrogen contacts<sup>19</sup> — suggests its use as a building block for single-molecule mechanical systems, where the six outer rings act as the interlocking teeth of a six-bladed molecular gear<sup>20</sup> (see ESI†). Substituted versions carrying different groups on each outer ring, made by varying the alkyne and tetraarylcyclopentadienone partners in the Diels–Alder/decarbonylation reaction used here, could in principle act as molecular rotors or switches whose positions can be controlled by external stimuli (heat, light, or chemical signals), with potential applications in molecular electronics, nanoscale information storage, and stimuli-responsive materials.

## Experimental

### General experimental methods

All reagents and solvents were obtained from commercial suppliers and used as supplied unless otherwise stated. Benzaldehyde was

freshly distilled prior to use. Melting points were determined on a Stuart SMP10 digital melting point apparatus and are uncorrected. Infrared spectra were recorded on a Bruker Alpha II FTIR spectrometer fitted with a diamond ATR module; absorptions are reported in  $\text{cm}^{-1}$ . The following abbreviations are used for the intensity of IR absorptions: s, strong; m, medium; w, weak; vs, very strong; br, broad; sh, shoulder.  $^1\text{H}$  NMR spectra were recorded on a Magritek Spinsolve 43 MHz benchtop spectrometer in deuterated chloroform ( $\text{CDCl}_3$ ) with tetramethylsilane ( $\text{Me}_4\text{Si}$ ) as internal standard; chemical shifts are reported in ppm ( $\delta$  scale) and coupling constants (J) are given in Hz. The following abbreviations are used for the multiplicities of NMR signals: s, singlet; d, doublet; t, triplet; m, multiplet; br, broad. Thin-layer chromatography was carried out on aluminium-backed Merck silica gel 60  $\text{F}_{254}$  plates and visualised under ultraviolet light at 254 nm.

### Benzoin

A solution of sodium cyanide in 95% ethanol (0.20 M, 5 mL) was added to freshly distilled benzaldehyde (1.0 mL, 1.06 g, 10.02 mmol) in a round-bottomed flask (10 mL) fitted with a magnetic stirrer bar and a reflux condenser. The stirred mixture was heated in a sand bath at 95–100 °C for 30 min, over which time the solution became yellow-orange and cloudy within approximately 5 min. After cooling to room temperature and then in an ice bath for 10 min, the precipitate was collected by suction filtration on a Hirsch funnel, washed with cold water ( $4 \times \text{mL}$ ) and air-dried under suction. The crude solid was recrystallised from 95% ethanol, washed with ice-cold 50:50 water–ethanol ( $4 \times 2 \text{ mL}$ ) and dried in vacuo to give benzoin (0.51 g, 48%) as colourless crystals. TLC analysis (silica,  $\text{CH}_2\text{Cl}_2$ ) of the recrystallised material showed a single spot at  $R_f$  0.63, while the filtrate showed a single spot at  $R_f$  0.50; m.p. 196.2 °C (from EtOH) (lit., 137 °C); IR (ATR)  $\nu_{\text{max}}/\text{cm}^{-1}$ : 3376 br (O–H), 3059 w (Ar C–H), 1677 s (C=O), 1594 m and 1448 m (Ar C=C), 753 s and 694 s (mono-substituted Ar C–H bend);  $\delta_{\text{H}}$  (43 MHz;  $\text{CDCl}_3$ ;  $\text{Me}_4\text{Si}$ ) 4.14 (1 H, br s, OH), 5.97 (1 H, s, CHOH) and 7.32–7.50 (8 H, m, ArH).

### Benzil

A mixture of benzoin (400 mg, 1.88 mmol) and cupric acetate catalyst solution (1.4 mL) was placed in a conical vial (5 mL) equipped with a magnetic spin vane and a reflux condenser (sealed at the joint with a rubber O-ring) fitted with a gas-delivery tube. A measuring cylinder (100 mL) filled with water was inverted into a beaker of water and clamped in position, with the outlet of the gas-delivery tube arranged beneath the cylinder mouth to collect evolved  $\text{N}_2$ . The stirred mixture was heated in a sand bath at 120–125 °C until nitrogen evolution ceased (approximately 2.5 h), during which the benzoin dissolved and the solution turned green (theoretical volume of  $\text{N}_2$  from 400 mg of benzoin = 42.4 mL at STP). After cooling to room temperature, cold water (2 mL) was added and the mixture was placed in an ice bath for 10 min. The yellow crystalline precipitate was collected by suction filtration on a Hirsch funnel and rinsed with cold water ( $2 \times 2 \text{ mL}$ ). TLC analysis of the crude material (silica,  $\text{CH}_2\text{Cl}_2$ , solvent front 4.0 cm) showed a major spot at  $R_f$  0.75 (benzil) together with a minor lower spot at  $R_f$  0.70 (residual benzoin); the filtrate displayed an elongated



streak around  $R_f \approx 0.50$  containing both compounds. The crude product was therefore purified by column chromatography (silica gel,  $\text{CH}_2\text{Cl}_2$ ) to give benzil (0.28 g, 71%) as yellow crystals; m.p. 91.3 °C (lit., 94–95 °C); IR (ATR)  $\nu_{\text{max}}/\text{cm}^{-1}$ : 3064 w (Ar C–H), 1657 s (C=O), 1592 m, 1490 w and 1449 m (Ar C=C), 716 s and 695 s (mono-substituted Ar C–H bend);  $\delta_{\text{H}}$ (43 MHz;  $\text{CDCl}_3$ ;  $\text{Me}_4\text{Si}$ ) 7.27–8.21 (10 H, m, ArH).

#### Tetraphenylcyclopentadienone

A mixture of benzil (100 mg, 0.48 mmol) and 1,3-diphenylacetone (100 mg, 0.48 mmol) in absolute ethanol (1.0 mL) was placed in a conical vial (3.0 mL) equipped with a magnetic spin vane and an air condenser. The mixture was heated with stirring in a sand bath at 90–100 °C for approximately 5–10 min until the reagents had dissolved. The vial was removed from the heat source and potassium hydroxide solution in ethanol (10% w/v, 200  $\mu\text{L}$ ) was added immediately with gentle swirling, whereupon dark purple crystals of the product began to form as the mixture cooled. Cold methanol (1.5 mL) was added with stirring and the mixture was placed in an ice bath for 5–10 min. The crystalline product was collected by suction filtration on a Hirsch funnel and washed with additional portions of ice-cold methanol ( $\geq 5$  mL total) until the brown impurity had been removed.

The solid was dried in vacuo to give tetraphenylcyclopentadienone (0.13 g, 70%) as deep purple crystals; m.p. 219 °C (lit., 220–221 °C); IR (ATR)  $\nu_{\text{max}}/\text{cm}^{-1}$ : 3397 br (O–H, residual solvent), 3063 w and 3029 w (Ar C–H), 1646 w (C=O), 1496 m and 1453 w (Ar C=C), 762 s and 688 s (mono-substituted Ar C–H bend);  $\delta_{\text{H}}$ (43 MHz;  $\text{CDCl}_3$ ;  $\text{CHCl}_3$ ) 6.82–7.75 (20 H, m, ArH).

#### (E)-Stilbene

Triacrylmethylammonium chloride (Aliquat 336, 300 mg) was placed in a round-bottomed flask (50 mL) equipped with a magnetic stirrer bar. Freshly distilled benzaldehyde (0.6 mL, 5.88 mmol), diethyl benzylphosphonate (2  $\times$  0.6 mL, 5.57 mmol), petroleum spirits (10 mL) and aqueous sodium hydroxide solution (40% w/v, 10 mL) were added, and the flask was fitted with a reflux condenser (lower joint lightly greased). The two-phase mixture was vigorously stirred and heated under reflux in a sand bath at 90–100 °C for 1 h, during which the solution developed an orange coloration. After cooling to room temperature, methylene chloride (4 mL) was added to dissolve any crystalline material that had formed, and the mixture was transferred to a separating funnel (125 mL); the flask was rinsed with further methylene chloride (2.0 – 3.0 mL) and the rinsings combined in the separating funnel. The aqueous layer was drained to an Erlenmeyer flask (50 mL), and the organic layer was washed with water (2  $\times$  6 mL); added to an Erlenmeyer flask (50 mL) and dried over anhydrous  $\text{Na}_2\text{SO}_4$ . The dried solution was decanted into another clean Erlenmeyer flask (50 mL), and the residual drying agent was rinsed with further methylene chloride (2  $\times$  3 mL); the rinses were combined with the organic solution. The combined organic filtrates were concentrated to dryness on a warm sand bath under a gentle stream of nitrogen. The crude solid was recrystallised from 95%

ethanol; the colourless crystals were collected by suction filtration on a Hirsch funnel, washed with ice-cold 50:50 water–ethanol and dried in vacuo for approximately 15 min. TLC analysis (silica,  $\text{CH}_2\text{Cl}_2$ ) of the recrystallised material showed a single spot at the solvent front ( $R_f$  1.0), while the filtrate displayed multiple spots distributed across the plate. The procedure gave (E)-stilbene (0.12 g, 12%) as colourless crystals; m.p. 121.8 °C (lit., 122–124 °C); IR (ATR)  $\nu_{\text{max}}/\text{cm}^{-1}$ : 3077 w, 3058 w and 3020 w (Ar/vinyl C–H), 1597 w and 1494 m (Ar C=C), 983 s (*trans* HC=CH out-of-plane bend), 761 s and 689 s (mono-substituted Ar C–H bend);  $\delta_{\text{H}}$ (43 MHz;  $\text{CDCl}_3$ ;  $\text{CHCl}_3$ ) 7.15 (2 H, s, CH=CH) and 7.24–7.72 (10 H, m, ArH).

#### meso-Stilbene dibromide

A mixture of (E)-stilbene (500 mg, 2.8 mmol) and glacial acetic acid (6 mL) was placed in a round-bottomed flask (50 mL) equipped with a magnetic stirrer bar and an air condenser (no water). The mixture was heated in a sand bath at 130–140 °C with stirring until the solid had dissolved (approximately 2–3 min). The condenser was briefly removed from the reaction flask in the fumehood and pyridinium bromide perbromide (1.0 g, 3.2 mmol) was added; any material adhering to the walls of the flask was washed down with further glacial acetic acid (6 mL), and the condenser was reattached. The mixture was stirred at 130–140 °C in a sand bath for further 5–6 min, during which the solution developed a bright orange coloration. The flask was removed from the heat and allowed to cool until warm to touch (~40–50 °C), whereupon colourless crystals of the product began to deposit. Water (10 mL) was added with swirling and the flask was placed in an ice bath for 5–8 min. The crystalline solid was collected by suction filtration on a Hirsch funnel, washed with cold water (4  $\times$  3 mL) followed by ice-cold acetone (3  $\times$  3 mL), and dried under vacuum to give meso-stilbene dibromide (0.98 g, 9%) as white crystals; m.p. 238 °C (lit., 241 °C, dec.); IR (ATR)  $\nu_{\text{max}}/\text{cm}^{-1}$ : 3054 w (Ar C–H), 1597 m, 1571 m, 1484 m and 1441 m (Ar C=C), 748 s, 716 s and 687 s (mono-substituted Ar C–H bend) and 568 m and 542 m (C–Br);  $\delta_{\text{H}}$ (43 MHz;  $\text{CDCl}_3$ ;  $\text{CHCl}_3$ ) 5.46–5.63 (2 H, s, CHBr) and 7.16–7.83 (10 H, m, ArH).

#### Diphenylacetylene

A mixture of meso-stilbene dibromide (400 mg, 1.2 mmol) and finely crushed potassium hydroxide flakes (400 mg, 22 mmol) was placed in an Erlenmeyer flask (10 mL) containing a magnetic stirrer bar. Triethylene glycol (2 mL) was added using a measuring cylinder, and the flask was placed in a preheated sand bath at 190–195 °C with vigorous stirring. The reaction was maintained at this temperature for approximately 10 min, during which the mixture developed a dark red-brown caramel coloration. The flask was removed from the heat and allowed to cool to approximately 40–50 °C (warm to the touch), at which point a fine powder was visible at the base of the flask. Distilled water (approx. 5 mL) was added with thorough stirring (giving a coffee-coloured suspension), and the mixture was placed in an ice bath for 10–15 min. The tan crystalline solid was collected by suction filtration on a Hirsch funnel under reduced pressure, washed with cold 70% ethanol (2  $\times$  2 mL), and dried in vacuo to give diphenylacetylene (0.19 g, 91%) as tan crystals; m.p. 60.4 °C (lit., 60–

61 °C); IR (ATR)  $\nu_{\text{max}}/\text{cm}^{-1}$ : 3063 w (Ar C–H), 1599 m, 1492 m and 1442 m (Ar C=C), 753 s and 687 s (mono-substituted Ar C–H bend);  $\delta_{\text{H}}$ (43 MHz;  $\text{CDCl}_3$ ;  $\text{CHCl}_3$ ) 6.95–7.92 (10 H, m, ArH).

### Hexaphenylbenzene

Tetraphenylcyclopentadienone (100 mg, 0.26 mmol) and diphenylacetylene (100 mg, 0.56 mmol) were placed in a Pyrex test tube (13 × 100 mm) and high-boiling silicone oil (1.5 mL) was added; the test tube was held at a slight angle, pointed away from the operator. The bottom of the test tube was gently heated above the flame of a microburner over a period of approximately 5 min, during which the reagents dissolved to give a dark red-purple solution. Heating was continued at a gentle boil for a further 10 min, over which time the deep purple colour faded and a fine tan crystalline solid began to deposit. The mixture was cooled to room temperature and petroleum spirits (4 mL) was added with stirring. The solid was collected by suction filtration on a Hirsch funnel and washed first with petroleum spirits (3 × 2 mL), then with cold toluene (2 × 2 mL), affording off-white crystals. The solid was dried in vacuo to give hexaphenylbenzene (0.076 g, 55%) as off-white crystals; m.p. not determined, exceeding the operating range of the available melting point apparatus (lit., 454–456 °C); IR (ATR)  $\nu_{\text{max}}/\text{cm}^{-1}$ : 3055 w and 3023 w (Ar C–H), 1599 w, 1577 w, 1494 m, and 1440 m (Ar C=C), 729 s and 694 s (mono-substituted Ar C–H bend);  $\delta_{\text{H}}$ (43 MHz;  $\text{CDCl}_3$ ;  $\text{CHCl}_3$ ) 6.20–7.62 (30 H, m, ArH).

### Conflicts of interest

There are no conflicts to declare

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